Abdelmalek Essadi University

Algebra IV — Normal Exam

Year: 2021

Exercise 1:

Let $a \in \mathbb{R}$. Consider the matrix

$$A_a = \begin{pmatrix} 2 & 2a-10 & -2 \\ -1 & a-1 & -1 \\ 0 & -3a+12 & 4 \end{pmatrix}.$$

1) 1) Determine $\mathrm{Rg}(A_a - I_n)$. Conclusion?
   2) For which values of $a \in \mathbb{R}$ is the matrix A_a diagonalizable?
2) 1) Write the minimal polynomial of A_a.
   2) Let $B = A_a$. Compute B^k for $k \in \mathbb{Z}$.
3) Choose $a = 1$ and set $A = A_1$.
   1) Determine a Jordan reduction of the matrix A.
   2) Let $P \in \mathbb{R}[X]$. Show that $P(A)$ is diagonalizable if and only if $P'(1) = 0$.

Answer Area

Exercise 2:

Let A be a nonzero matrix in $M_3(\mathbb{R})$ satisfying $A^3 = -A$.

1) Determine the characteristic polynomial of A.
2) Let $A = M(f, B_3)$. Establish the relation

$$\mathbb{R}^3 = \ker f \oplus \ker(f^2 + \mathrm{id}_{\mathbb{R}^3}).$$

3) * Conclude that the matrix A is similar to

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Answer Area